

Q : What is priority queue? Discuss the one way list representation of a priority queue. Compare the one way list and array representation of priority queues.

Priority queue : A priority queue is a queue in which elements are assigned with priority and deleted as the two rules.

- 1.The element with higher priority is deleted before any element of lower priority.
- 2.The elements with the same priority are deleted according to the order in which they were added to the queue.

One way list representation of a priority queue :

One way list representation of a priority queue works as the two rules.

- 1.Each node in the list must contain three information.

(a)Data (INFO)

(b)Priority number (PRN)

(c)Memory address (LINK)

2.A node X preceeds a node Y in the list.

(a)When X has higher priority than Y.

(b)When X and Y have same priority but X is added in the list before Y.

Comparison between one way list and array representation of priority queues :

1.An array is a collection of elements while a linked list is a collection of objects called nodes that contain data and addresses.

2.Array representation of a priority queue requires two information: data and priority number . On the other hand one way list representation of a priority queue requires three information: data,priority number and address.

Q : What is linked list? What are the advantages and disadvantages of linked list over the linear array.

Linked list : A linked list is a linear collection of data elements called nodes where each node points the location of the next node in the list.

Advantages of linked list over the linear array :

1. Linked lists has dynamic size while array has fixed size.
2. Linked list has no wasted memory while array has wasted memory.

Disadvantages of linked list over the linear array :

1. Linked list accesses the elements slow comparatively while array accesses the elements quick.
2. Linked list requires extra memory for the pointers while array does not use pointers.

Q : What is circular header list? Write the advantages of circular header list over ordinary linked list.

Circular header list : A circular header list is a header list where the last node points back to the header node.

Advantages of circular header list over ordinary linked list :

- 1.The null pointer is not used in circular header list. Hence it contains a valid address.
- 2.Each node of circular header list has a predecessor. So the first node does not require a special node like ordinary linked list.

Q : What is two way linked list ?

Two way linked list : Two way linked list is a linear collection of data elements called nodes where each node has two pointers.

- 1.One points the location of the next node in the list.
- 2.Another points the location of the preceding node in the list.

Q : Write an algorithm to sort an array A of n elements using insertion sort.

Step 1 : Start

Step 2 : Set $I:=2$

Step 3 : Repeat step-4 to step-7 while $I \leq N$

Step 4 : Set $temp:=DATA[I]$ & $J:=I-1$

Step 5 : Repeat step-6 while $J \geq 1$ & $DATA[J] > temp$

Step 6 : Set $DATA[J+1]:=DATA[J]$ & $J:=J-1$

Step 7 : Set $DATA[J+1]:=temp$

Step 8 : Set $I:=I+1$

Step 9 : End

Q : Write an algorithm to sort an array A of n elements using selection sort.

Step 1 : Start

Step 2 : Set I:=1

Step 3 : Repeat step-4 to step-9 while I<N

Step 4 : Set MIN:=I & J:=I+1

Step 5 : Repeat step-6 to step-7 while J<=N

Step 6 : If DATA[J] < DATA[MIN]

Set MIN:=J

Step 7 : Set J:=J+1

Step 8 : Swap DATA[MIN] & DATA[I]

Step 9 : Set I:=I+1

Step 10 : End

Q : What is binary search tree?

Binary search tree : Binary search tree is a binary tree in which the value of left child is less than the value of its root and the value of right child is greater than the value of its root.

Q : What is max heap?

Max list : Max heap is a binary tree in which the value of the root is greater than its two children.

Q : Consider a situation where swap operation is very costly. Which sorting algorithms should be preferred among heap, selection, insertion and merge so that the number of swap operations are minimized in general? Why?

Selection sort should be preferred. Because selection sort performs minimum swap operations among them.

Selection sort makes $O(n)$ swap operations.

Q : Explain which sorting algorithms is not stable in its typical implementation among insertion,merge,bubble and quick.

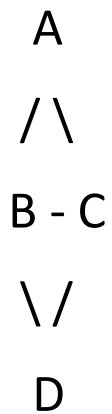
Quick sort is an unstable sort. Because in this sort swapping of elements is according to pivot's position without considering their original positions.

Q : Which data structure is most efficient to find the top 10 largest items out of 1 million items sorted in file among min heap, max heap, BST, sorted array ? Why ?

Max heap is the most efficient data structure among them. Because in max heap the largest number is always at the root.

Q : Describe the adjacency list and linked representation of graph with necessary figure.

Figure :



Adjacency list :

A: B, C

B: A, C, D

C: A, B, D

D: B, C

- 1.Vertex A is adjacent to vertices B and C.
- 2.Vertex B is adjacent to vertices A, C, and D.
- 3.Vertex C is adjacent to vertices A, B, and D.
- 4.Vertex D is adjacent to vertices B and C.

Linked representation :

A -> B -> C

B -> A -> C -> D

C -> A -> B -> D

D -> B -> C

- 1.Vertex A is connected to vertices B and C.
- 2.Vertex B is connected to vertices A, C, and D.
- 3.Vertex C is connected to vertices A, B, and D.
- 4.Vertex D is connected to vertices B and C.

Solved by Mr. Pirate-3168 (3EM)

